CARTESIAN SQUARES OF LIMITS ARE DETECTED ON THE SYSTEMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5534-5560, following the statement at 5499-5532.

Editorial note. The first half's 'it is immediate from the construction of X^1_i' is a construction-reading step anchored to the remark and the defining scheme-theoretic images; flagged in the notes.

Allowed standard prerequisites (editorial).

- Well-definedness of the redefined system. remark-finite-type-gives-well-defined-system (same chapter, cited at ~5535-5539). The choice of W^1 is immaterial; take W^1 = W^2 x_{W^4} W^3.
- Morphisms of limits over good diagrams. lemma-morphism-good-diagram (same chapter, card L26309, cited at ~5518-5522). Supplies the system morphisms a_i, b_i, p_i, q_i.
- Flat finite presentation morphisms are constant on the system. lemma-morphism-good-diagram-flat (same chapter, card L26321, cited at ~5549-5551). Applied to a, b, p, q and their composites in the flat case.
- Descent of isomorphisms. lemma-descend-isomorphism (same chapter, cited at ~5558-5559). (q_i, p_i) is an isomorphism for i large.

Source reference. Original equation 0CP8: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5264–5264. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0852: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1925–1925. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CPA: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5324–5324. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CPC: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5424–5424. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 0CPB: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5404–5404. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CP6: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5249–5249. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0CP6 suppose that we have a cartesian diagram*

$$\begin{array}{ccc} X^1 & \xrightarrow{p} & X^3 \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle a} \\ X^2 & \xrightarrow{b} & X^4 \end{array}$$

of algebraic spaces quasi-separated and of finite type over B. For each $j = 1, 2, 3, 4$ choose $i_j \in I$ and a diagram

$$\begin{array}{ccc} X^j & \longrightarrow & W^j \\ \downarrow & & \downarrow \\ B & \longrightarrow & B_{i_j} \end{array}$$

as in (0CP8). Let $X^j = \lim_{i \geq i_j} X^j_i$ be the corresponding limit descriptions as in Lemma 0CPA. Let $(a_i)_{i \geq i_5}$, $(b_i)_{i \geq i_6}$, $(p_i)_{i \geq i_7}$, and $(q_i)_{i \geq i_8}$ be the corresponding morphisms of inverse systems constructed in Lemma 0CPA. Then there exists an $i_9 \geq \max(i_5, i_6, i_7, i_8)$ such that for $i \geq i_9$ we have $a_i \circ p_i = b_i \circ q_i$ and such that

$$(q_i, p_i) : X^1_i \longrightarrow X^2_i \times_{b_i, X^4_i, a_i} X^3_i$$

is a closed immersion. If a and b are flat and of finite presentation, then there exists an $i_{10} \geq \max(i_5, i_6, i_7, i_8, i_9)$ such that for $i \geq i_{10}$ the last displayed morphism is an isomorphism.

Proof. According to the discussion in Remark 0CPB the choice of W^1 fitting into a diagram as in (0CP8) is immaterial for the truth of the lemma. Thus we may choose $W^1 = W^2 \times_{W^4} W^3$. Then it is immediate from the construction of X^1_i that $a_i \circ p_i = b_i \circ q_i$ and that

$$(q_i, p_i) : X^1_i \longrightarrow X^2_i \times_{b_i, X^4_i, a_i} X^3_i$$

is a closed immersion.

If a and b are flat and of finite presentation, then so are p and q as base changes of a and b. Thus we can apply Lemma 0CPC to each of a, b, p, q, and $a \circ p = b \circ q$. It follows that there exists an $i_9 \in I$ such that

$$(q_i, p_i) : X^1_i \to X^2_i \times_{X^4_i} X^3_i$$

is the base change of (q_{i_9}, p_{i_9}) by the morphism by the morphism $X_i^4 \to X_{i_9}^4$ for all $i \geq i_9$. We conclude that (q_i, p_i) is an isomorphism for all sufficiently large i by Lemma 0852. $\square$